Ashif Ahmed Shuvo

Agrometeorologist & Food Engineer

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About me

I am a results-driven researcher and AI specialist with a B.Sc. in **Food Engineering** and an M.Sc. in **Agrometeorology** from Bangladesh Agricultural University. I specialize in developing high-precision, lightweight deep learning frameworks, such as **EinsteinNet, HEART-Net,** and **HazardNet** (*manuscript under preparation*). My expertise bridges **computer vision, IoT-driven automation,** and **spatio-temporal modeling** to address real-world agricultural challenges and climate resilience.

Technical Skills

Languages: Python, R, Java, C, C++
Computer Vision: ***HazardNet, HEART-Net, EinsteinNet***, Vision Transformer, Swin Transformer, DeepLabV3+, ResNet, DenseNet, EfficientNet, U-net, FCN, Attention Mechanism
Packages: TensorFlow, RDKit, OpenCV, Keras API, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Data Analysis: Cluster analysis, Regression, PCA, PLS, Multivariate Analysis
Cloud Computing: Microsoft Azure, AWS, Cloudflare
Spectroscopy: NIR, Spectrophotometer
Design Tools: Blender

Education

Bangladesh Agricultural University, Mymensingh. March 2025 - Oct 2026
M.Sc. in Agrometeorology
Bangladesh Agricultural University, Mymensingh. Jan 2018 - Oct 2024
B.Sc. in Food Engineering
Government M. M. College, Jashore. June 2014 - April 2016
Higher Secondary Certificate (HSC) in Science
Jashore Zilla School, Jashore. Jan 2007 - April 2014
Secondary School Certificate (SSC) in Science

Publications & Datasets

- **Shuvo, Ashif Ahmed,** Hasan, Sakibul, Anjum, Abir,Shahed, Asaduzzaman, Al-Mamun, Mohammad, *HEART-Net: An ultra-lightweight multimodal neural network for on-device cattle live-weight estimation and precision phytogenic dosing* (2026). **Smart Agricultural Technology**. https://doi.org/10.1016/j.atech.2026.102587
- **Shuvo, Ashif Ahmed**, Bhuian, Wahada Jinnat Oishy, Rahman, Afzal, Iqbal, Abdullah, *EinsteinNet and State-of-the-Art ML Models for Android-Based Orange Classification: Integration, Evaluation, and Deployment* (2025). **Smart Agricultural Technology**. https://doi.org/10.1016/j.atech.2025.101072
- **Shuvo, Ashif Ahmed;** Monika, Most.Subrina Afrin; Bhuian, Wahada Jinnat Oishy, *Orange Fruit Image Dataset for Classification* (2026), Mendeley Data. https://doi.org/10.17632/zk4hzhtcdd.2

- **Shuvo, Ashif Ahmed**, *A Comprehensive Retail Market Price Dataset for Wheat Flour and Macroeconomic Indicators in Bangladesh* (2026). Mendeley Data. https://doi.org/10.17632/5m4d33mxrs.1

Projects

HazardNet (*https://hazardnet.live*) Sep 2026 - Present

- Multi-hazard early warning for Bangladesh agriculture. HazardNet publishes dated 7- and 15-day hazard outlooks for all 64 districts of Bangladesh, including eight hazard classes, a severity value and a confidence score per district and forecast date, in two honestly labelled severity tracks.

ResearchWithME (*https://catalystlab.tech*) June 2026 - Present

- Eight diagnostic engines run in parallel on one domain and return a scored report with the measurement behind every field.

CatalystLab (*https://researchwith.me*) May 2025 - Present

- Eight diagnostic engines run in parallel on one domain and return a scored report with the measurement behind every field.

Experiences

Research Assistant, Bangladesh Agricultural University Research System Mar 2025 - Present

- Developed HEART-Net, a Green AI and privacy-aware hybrid cognitive architecture designed for real-time cattle weight estimation across decentralized agricultural divisions in Bangladesh.

Internship, Data Analytics and Visualization Job Simulation, Accenture Nov 2024

- Successfully completed practical tasks in Project Understanding, Data Cleaning & Modeling, Data Visualization & Storytelling, Present to the Client **(*Certificate of Completion*)**.

Diploma, Python Development Essentials, Institute of MTF Oct 2024 - Nov 2024

- Gained proficiency in data structures, functions, and OOP **(*Certificate of Completion*)**.

Internship, Data Analytics, Quantium Sep 2024 - Nov 2024

- Analyzed transaction data to identify benchmark stores and deliver insights for strategic commercial decisions **(*Certificate of Completion*)**.

Internship, Akij Food and Beverages Ltd., Bangladesh. June 2024 - July 2024

- Completed a 15-day internship at AFBL focused on mango pulp, chutney, and mustard oil production, emphasizing HACCP, quality control, and logistics.

Achievements

Bangladesh Government Merit Scholarship at **JSC, SSC, & HSC** exams.

Professional Development & Online Learning

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1. **OpenCV Bootcamp** from ***OpenCV University*** (Grade Earned - 100%)**.**
2. **TensorFlow Keras Bootcamp** from ***OpenCV University*** (Grade Earned - 100%)**.**
3. **AI Agents Fundamentals** from ***Hugging Face*****.**
4. **Intro to Programming** from ***Kaggle*****.**
5. **Python** from ***Kaggle*****.**
6. **Machine Learning Explainability** from ***Kaggle*****.**
7. **Machine Learning Online Course** from MakeIntern through ***Udemy*****.**
8. **Data Science Career Path** from MakeIntern through ***Udemy*****.**
9. **Internet of Things (IoT) Online Course** from MakeIntern through ***Udemy*****.**
10. **Python for OOP: The A-to-Z OOP Python Programming Course** from Meta Brains through ***Udemy*****.**
11. **Python & TensorFlow: Deep Dive into Machine Learning** from Meta Brains through ***Udemy*****.**
12. **Python for Deep Learning: Build Neural Networks in Python** from Meta Brains through ***Udemy*****.**

13. **Harnessing AI for a social impact** from ***Udemy***.
14. **Harnessing AI and Machine Learning for Geospatial Analysis** from ***Udemy***.

Leadership & Extracurricular Activities

Vice President, Nutrition Club, BAU — Apr 2023 - Present
- Lead 2 national-level competition & arranged more than 20 workshops and seminars

IT Advisor, Shwapno Youth Development Organization — July 2020 - Nov 2021
- Provided emergency support during COVID-19 pandemic & organized around 15 training sessions

Campus Ambassador, English Olympiad — Jan 2019 - Dec 2020
- Co-ordinated teams from 40 educational institutions & arranged competition

General Member, Volunteer for Bangladesh. — Feb 2019 - Oct 2020
- Fundraising for unprivileged children
- Several social activities and awareness problems

Academic Reference

Dr. Ahmed Khairul Hasan
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Bangladesh Agricultural University, Mymensingh.
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Dr. Mohammad Jahangir Alam
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